Supplementary Information

Advanced Space-Charge-Limited Current Model for Analyzing Fermi Level Shift in the Bandgap of Halide Perovskites

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Supplementary structural and optical characterization of $MAPbI_3$ and $MAPbBr_3$.

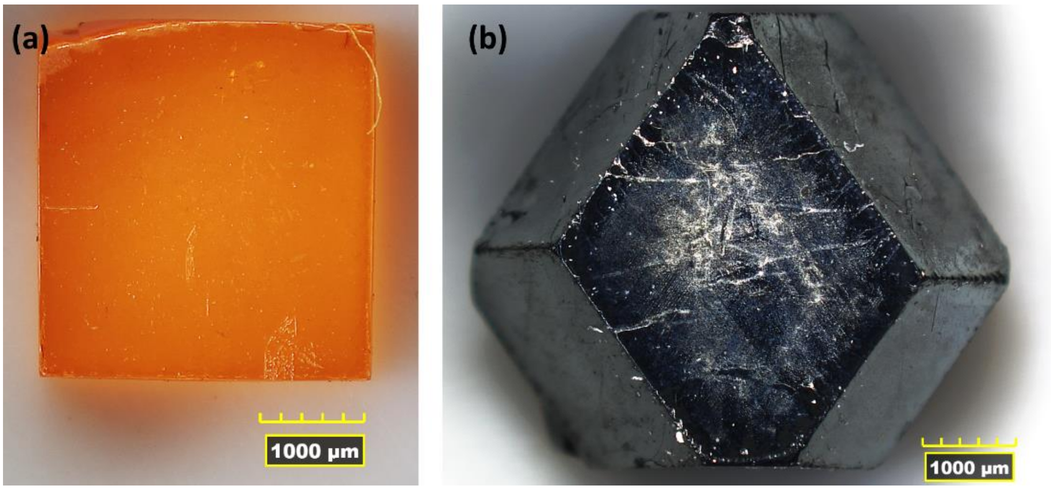


Fig. S1 Microscopic images of $MAPbBr_3$ (a) and $MAPbI_3$ (b) single crystals taken by HIROX RH-2000 digital microscope (magnitude 35x).

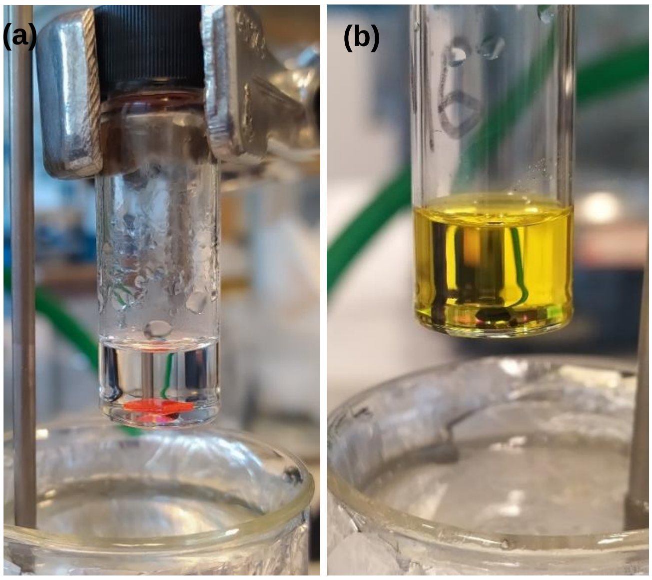


Fig. S2 Crystals of (a) $MAPbBr_3$ and (b) $MAPbI_3$ growing in polar aprotic solutions.

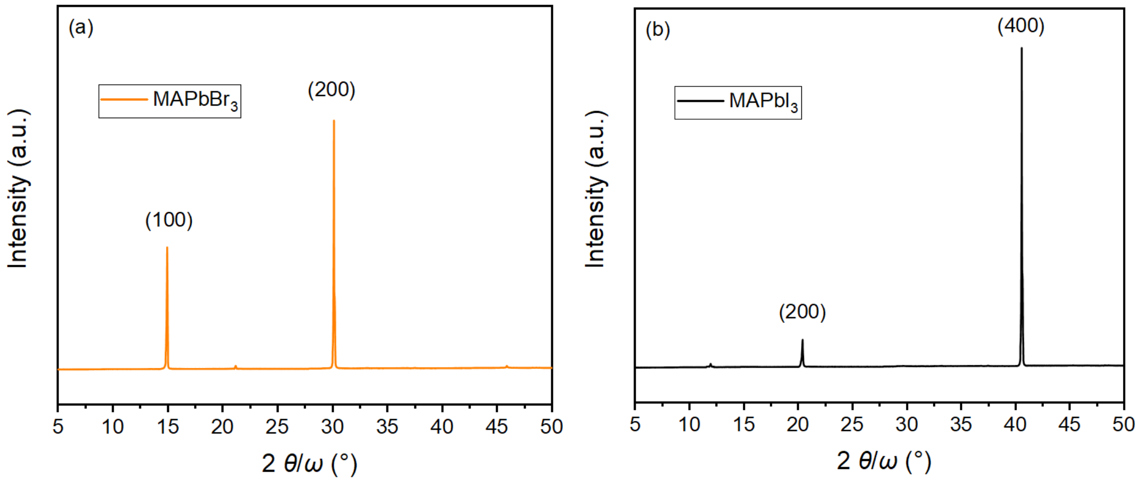


Fig. S3 XRD patterns of (a) $MAPbBr_3$ and (b) $MAPbI_3$.

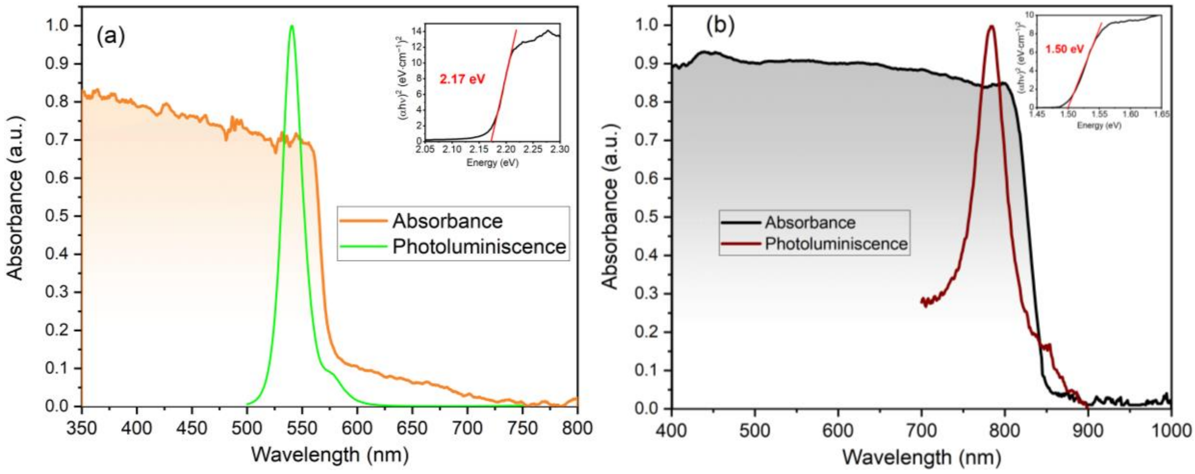


Fig. S4 Absorbance and photoluminescence spectra of (a) $MAPbBr_3$ and (b) $MAPbI_3$.

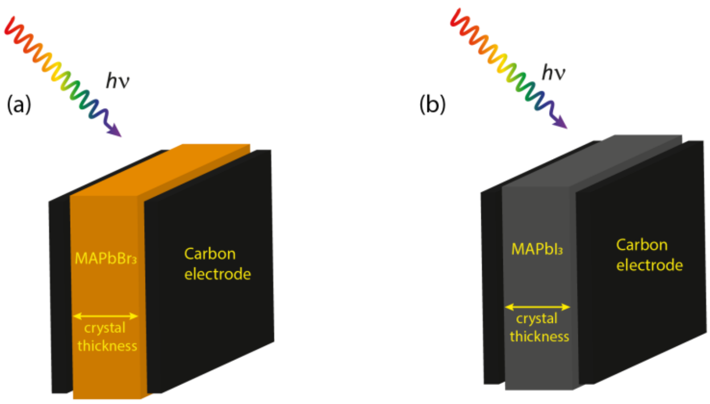


Fig. S5 Schemes of the studied samples: (a) C/$MAPbBr_3$/C and (b) C/$MAPbI_3$/C.

Table S1 Parameters of the prepared samples.

Sample	Thickness (mm)	Active area (mm^2)
$MAPbBr_3$	0.80	7.70
$MAPbI_3$	1.05	20.10

Supplementary Note 1: Determination of density of states distribution.

A model consisting of a square root (approximately monoenergetic) transport band and one bi-exponential (approximately monoenergetic) trap state was used to describe the experimental dependences of the free and trapped charge carrier concentrations of the studied perovskites. Energy distribution of states in conduction (for $E \geq E_c$), and valence band (for $E \leq E_v$) can be described by the following equations:

$$g(E) = \frac{8\pi m_e^*}{h^3} \sqrt{2m_e^*(E - E_c)}\,, \tag{S1}$$

$$g(E) = \frac{8\pi m_h^*}{h^3} \sqrt{2m_h^*(E_v - E)}\,, \tag{S2}$$

where m_e^* and m_h^* are the effective masses of free electrons and holes (here, for p-type $MAPbBr_3$ m_h^* = 0.305 and 0.35 for $MAPbI_3$[1],[2]). For non-degenerate semiconductors, in which the Fermi level lies at a maximum distance of $3k_BT$ from the conduction edge (E_c), or of the valence edge (E_v) band, the parabolic bands can be replaced by a mono-energy level of concentration[3]

$$N_c = \frac{4\pi m_e^*(k_BT)}{h^3} \sqrt{2\pi m_e^*(k_BT)}\,, \tag{S3}$$

$$N_v = \frac{4\pi m_h^*(k_BT)}{h^3} \sqrt{2\pi m_h^*(k_BT)}\,. \tag{S4}$$

The trap states (intrinsic and extrinsic defects, vacancies)[4] typically present in the bandgap of halide perovskite semiconductors can be repsesented by a Gaussian or biexponential distribution of states $g(E)$,

$$g(E) = \frac{N_t\, e^{(E-E_t)/k_BT_t}}{k_BT_t\left[1 + e^{(E-E_t)/k_BT_t}\right]^2}\,, \tag{S5}$$

$$g(E) = \frac{N_t}{\sigma\sqrt{2\pi}}\, e^{(E-E_t)^2/2\sigma^2}\,, \tag{S6}$$

where N_t is concentration of localized states, E_t is energy position of distribution maximum (trap), $\sigma = 2k_BT_t$ is the standard deviation from average value, with its characteristic trap temperature, T_t.

For the mentioned model distributions, the concentration of free n_f (p_f) and trapped charge carriers n_t (p_t) in the band gap of the semiconductor can be determined using convolution integrals

$$n_f(E_F) = \int_{Ec}^{\infty} g(E)\, f(E - E_F)\mathrm{d}E\,, \tag{S7}$$

$$n_t(E_F) = n_s(E_F) - n_f(E_F) = \int_{E_{F0}}^{E_c} g(E)\, f(E - E_F)\mathrm{d}E\,, \tag{S8}$$

where $n_{\mathrm{s}} = n_{\mathrm{t}} + n_{\mathrm{f}}$ is the sum of free and trapped charge concentrations. The Fermi-Dirac distribution function $f(E - E_{\mathrm{F}})$ is defined as

$$f(E_{\mathrm{F}} - E) = \frac{1}{1 + \exp\left[(E - E_{\mathrm{F}})/k_{\mathrm{B}}T\right]}. \tag{S9}$$

where E_{F} is the Fermi energy. Equation (S7) and (S8) can be written for positive charge (e.g. for holes) using following relations

$$p_{\mathrm{f}}(E_{\mathrm{F}}) = \int_{-\infty}^{Ev} g(E)\left[1 - f(E - E_{\mathrm{F}})\right] \mathrm{d}E, \tag{S10}$$

$$p_{\mathrm{t}}(E_{\mathrm{F}}) = p_{\mathrm{s}}(E_{\mathrm{F}}) - p_{\mathrm{f}}(E_{\mathrm{F}}) = \int_{E_{\mathrm{v}}}^{E_{\mathrm{F0}}} g(E)\left[1 - f(E - E_{\mathrm{F}})\right] \mathrm{d}E. \tag{S11}$$

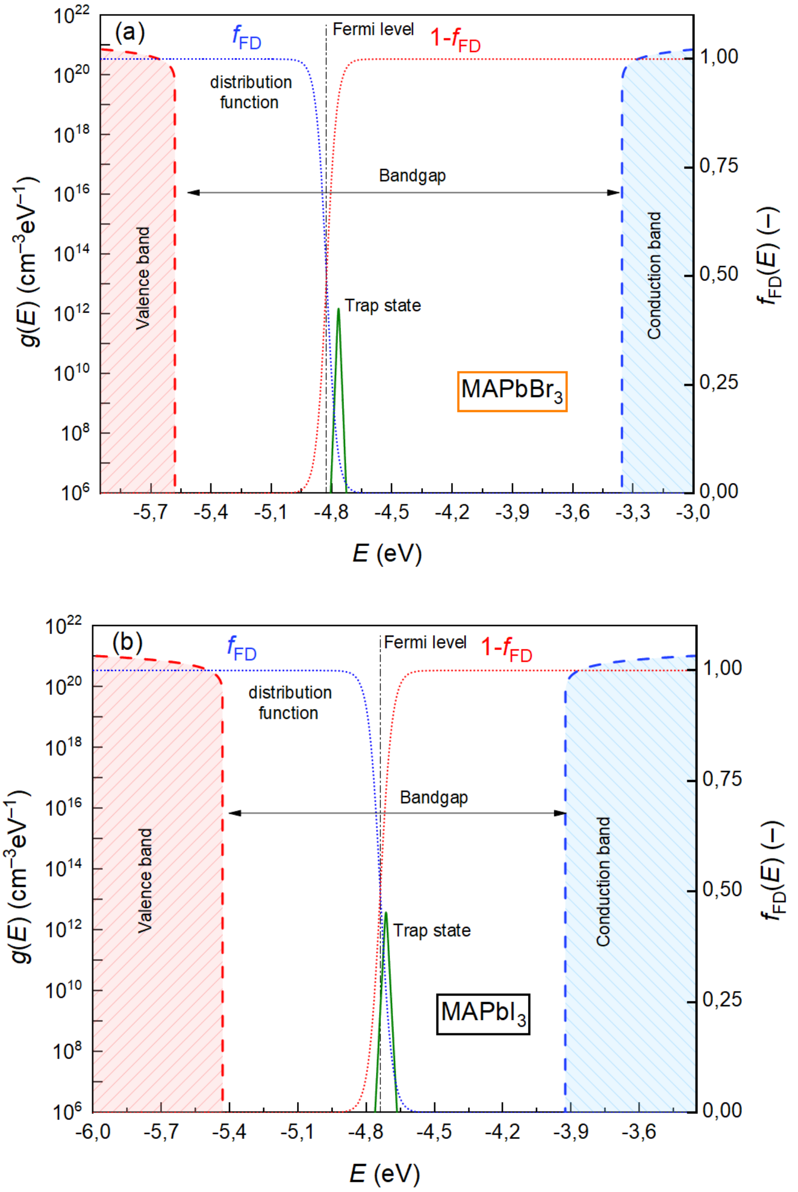


Fig. S6 Density of state distribution, Fermi-Dirac distribution functions for electrons (f_{FD}) and holes ($1–f_{\mathrm{FD}}$), and trap state used for modeling of (a) $MAPbBr_3$ and (b) $MAPbI_3$ halide perovskites.

Table S2 All parameters used to build the A-SCLC model.

Parameter	Quantity (Unit)	
Voltage	V (V)	Measured
Current density	J (Am^{-2})	
Temperature	T (K)	
Conduction band	E_c (eV)	Fixed (Constants)
Valence band	E_v (eV)	
Intrinsic Fermi level	$E_{F0,in}$ (eV)	
Bandgap	E_g (eV)	
Dielectric constant	ε_r (–)	
Electron and hole effective mass	m_e^*, m_h^*(g)	
Concentration of delocalized states	N_v, N_c (m^{-3})	
Sample thickness	L (m)	
Active area	S (m^2)	
Other physical constants	k_B (JK^{-1}), h (Js) e(C), ε_0 (Fm^{-1})	
Microscopic mobility	μ_0 ($m^2V^{-1}s^{-1}$)	**Variable**
Thermodynamic Fermi level	E_{F0} (eV)	
Concentration of localized states	N_t (m^{-3})	
Trap position	E_t (eV)	
Trap temperature	T_t (K)	
Current density voltage function	$J = f(V)$ (Am^{-2})	Modeled / Calculated
Reverse slope of the *J*-*V* curve	γ (–)	
Effective mobility	μ_{eff} ($m^2V^{-1}s^{-1}$)	
Free and trapped charge carrier concentrations	p_f, n_f (m^{-3}) p_t, n_t (m^{-3})	
Fermi level position	E_F (eV)	
Fraction of free and total charge carrier concentration	Θ (–)	

Supplementary modeling scheme.

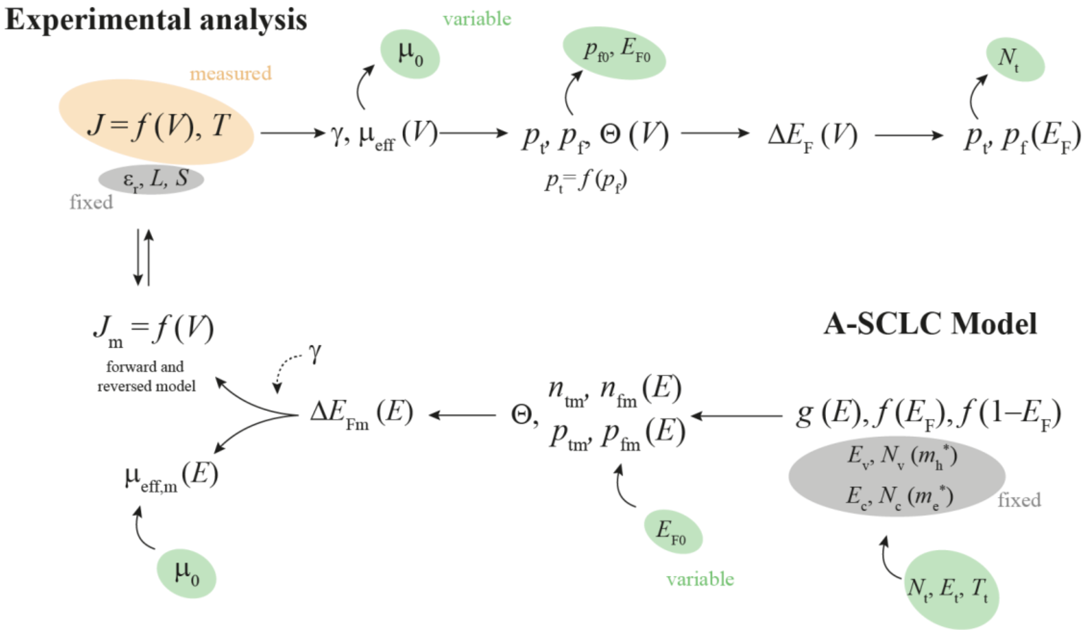


Fig. S7 A scheme describing analysis of experimental data and A-SCLC modeling approach.

Supplementary Note 2: A guide (see Fig. S7) for *J-V* curve analysis (A) and modeling (M).

A1. Measure the steady-state *J-V* at constant temperature.

A2. Calculate the slope (m) of the logarithmic dependence of the *J-V* curve according to the following equation $\gamma = 1/m = d\ln V / d\ln J$.

A3. Calculate the effective mobility (μ_{eff}) from equation (4) (see the main text). If the dependence plots a U-shape curve, then determine/estimate the microscopic mobility μ_0. *Note: the value of the microscopic mobility can also be estimated from the Ohmic region, where $p_t = p_f$.*

A4. Calculate the trapped (p_t) and free (p_f) charge carrier concentrations according to equations (5) and (6) described in the main text. If the p_t equals p_f (usually occurs at higher voltages), the charge starts to occupy the transport band, and the Mott-Gurney's law applies. Determine the parameter theta using equation $\Theta = p_f / p_s = p_f / (p_f + p_t)$.

A5. Calculate the thermodynamic Fermi level E_{F0}, and then the Fermi level shift for a given temperature according to equation (7) from the main text.

A6. Plot the dependence of the concentration of trapped carriers on the concentration of the free charge carriers ($p_t = f(p_f)$). Also, plot these concentrations as a function of the Fermi energy position in the bandgap ($p_t = f(E_F)$ and $p_f = f(E_F)$) of the given semiconductor (energy band diagram). From the saturation of trapped charge carriers, determine the value of N_t (see Fig. S8b,c).

A7. By comparing the E_{F0} value, the energy distribution of charge carriers, and the output work functions of the contacts, determine whether it is a p- or n-type semiconductor.

M1. Based on the semiconductor parameters, calculate the density of states $g(E)$ and the Fermi distribution functions for holes and electrons (using equations S1–S11). In the calculation, include the density of states of a monoenergetic trap (this case), or alternatively another distribution of states (Gaussian or biexponential distribution of states). If necessary, add additional trap states. Enter the values of N_t, E_t and T_t: use N_t based on the analysis of experimental data, the exact value of E_t is determined by modeling the parameters and subsequent optimization, the value of T_t has no significant effect on the shape of the trapped charge carriers (see Fig. S12).

M2. Using the convolution integral, determine the concentration of trapped and free charge carriers for further modeling of the *J-V* characteristic (in the calculation, include the E_{F0} value obtained from the analysis of experimental data), determine the theta parameter of the model being created.

M3. Determine the Fermi level shift for the given temperature.

M4. Finally, calculate the current density J_{mod} as a function of the parameter γ. Since the value of $\gamma \in (0;1)$ for charge injection (more precisely, $\gamma \in (0;\ 0.5\rangle$), this parameter does not fundamentally affect the model of the *J-V* characteristic. The following equation applies: $\gamma = \dfrac{T}{T_t + T}\ (T_t \geq T),\ \gamma = 0.5\ (T_t < T)$.

M5. Determine the effective mobility (optimize the value of the microscopic mobility μ_0) for the model data.

Supplementary Note 3: Model simulations.

To gain a better understanding of the modeled parameters and their correlations we performed multiple model simulations (Figs. S8–S12). Firstly, we simulated how model curves of trapped charge carriers (p_{tm} and n_{tm}) change with the position of thermodynamic Fermi level (E_{F0}). Fig S8a shows three possible positions of the thermodynamic Fermi level relative to the fixed position of the trap state: (1) Fermi level closer to the conduction band ($E_{F0} > E_t$); (2) Fermi level at the position of the trap state ($E_{F0} = E_t$); (3) Fermi level closer to the valence band ($E_t > E_{F0}$). The position (1) is considered as the shallow trap state for holes (deep trap state for electrons), while the position (3) presents the opposite situation, the deep trap state for holes and the shallow trap state for electrons. The situation at position (2) presents the deep trap state for both holes and electrons. To clarify the relationship between p_{tm} (n_{tm}) and *J-V* curves we marked the individual regions (ohmic, trap, TFL and Mott-Gurney) for situation (1) and (3). In case of monoenergetic shallow trap state, the current is firstly proportional to voltage ($J \sim V^1$), while the filling of the trap state introduces the space charge (i.e. nonlinear *J-V* characteristics) in the given semiconductor. The trap itself follows $J \sim V^2$, after which the current enters the trap filled limit region ($J \sim V^n$), where it is rapidly filled with charge carriers. Then the charge carriers start occupying the transport band where the delocalized charge transport takes place (i.e. Mott-Gurney law applies and $J \sim V^2$). The total concentration of the trapped charge carriers (p_{tm}) can be determined from the place where the p_{tm} hits the saturation value (see Fig. S8b,c). In the case of the deep trap state where $E_{F0} \approx E_t$ or $E_{F0} = E_t$, the *J-V* curves change from ohmic directly to TFL region. The lastly described situation (2) is the typical example of *J-V* curves of halide perovskite-based devices reported in the literature[5]–[10].

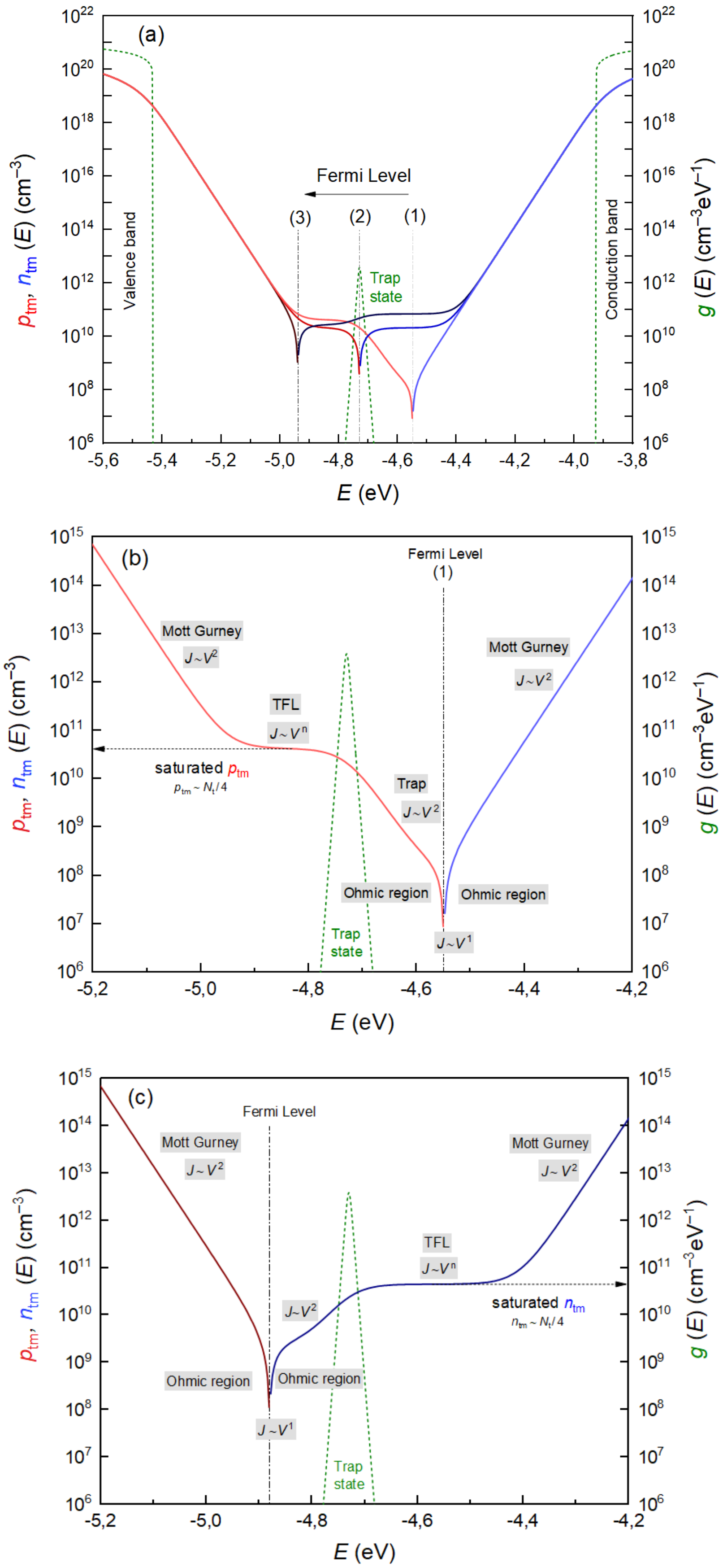


Fig. S8 (a) p_t and n_t for three different Fermi Level positions: (1) closer to the conduction band, (2) at the trap position, (3) closer to the valence band; (c,d) isolated examples of situations (2) and (3) with corresponding regions.

Furthermore, the energy position of monoenergetic trap state also affects the shape of p_{tm} and n_{tm} curves (Fig. S9). Let us consider the situation (2) where thermodynamic Fermi level is fixed at the position of the trap state ($E_t = E_{F0} = -4.70$ eV), and simulate how modeled curves p_{tm} and n_{tm} change relative to the E_t position.

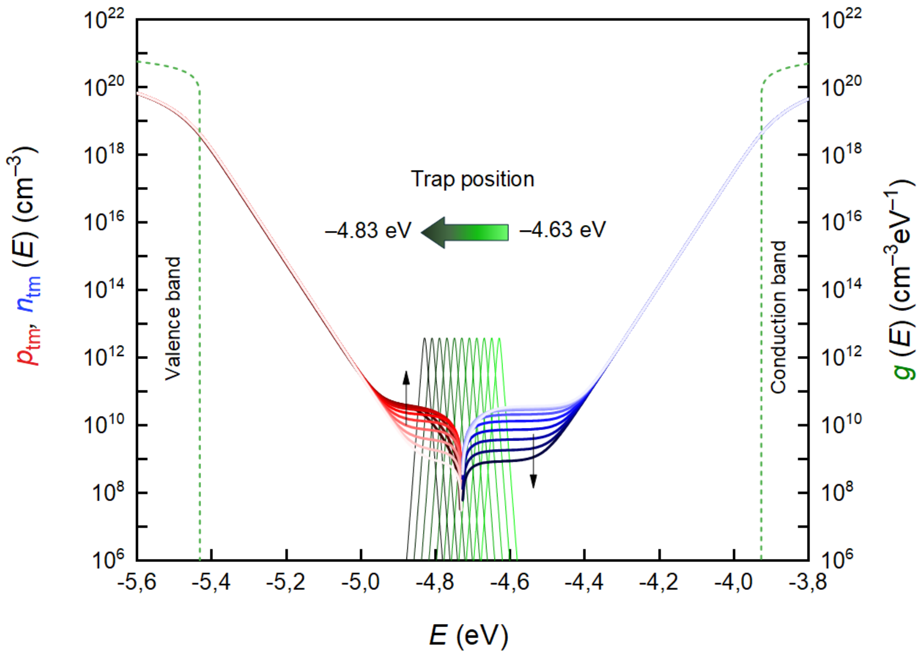


Fig. S9 Change of p_{tm} and n_{tm} with variable trap position.

Two temperatures affect the proposed model: (i) the temperature T (K) at which the experiment was carried out, and (ii) the trap state temperature T_t (K), which describes monoenergetic nature of the trap. Fig. S10 and Fig. S11 show the effect of T (K) on the p_{tm}, n_{tm} and the increment of hole concentration p_{tm}/dE_F. As evident, the mapping of the trap state and its effect on other parameters becomes more significant for lower temperatures. For higher temperatures (T>350 K) the trap is not visible in the p_{tm}, n_{tm} curves.

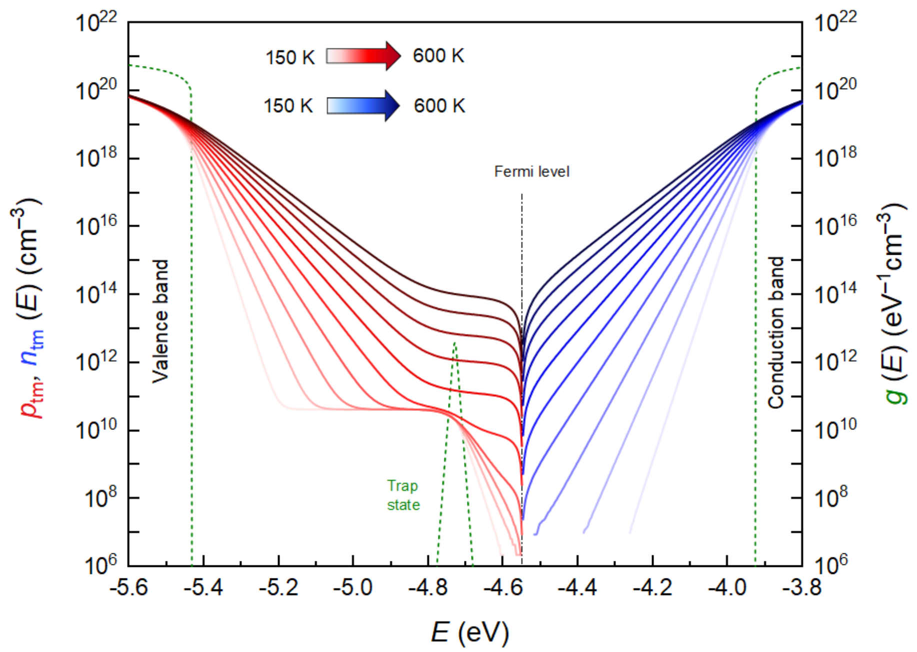


Fig. S10 The effect of temperature T (K) on p_{tm} and n_{tm} in the range of 150–600 K (50 K step)

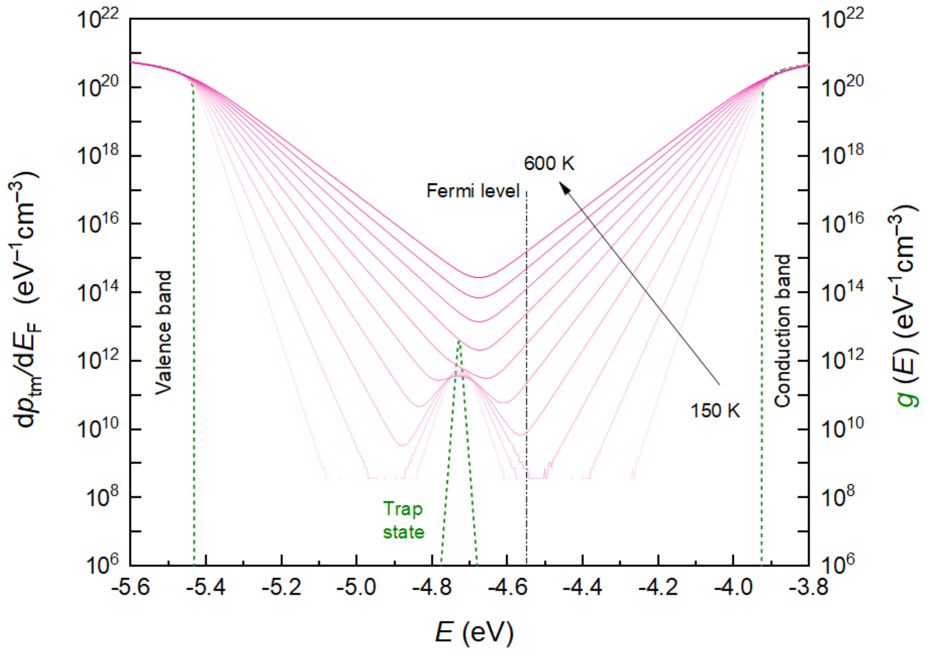


Fig. S11 The effect of temperature T (K) on dp_{tm}/dE_F in the range of 150–600 K (50 K step).

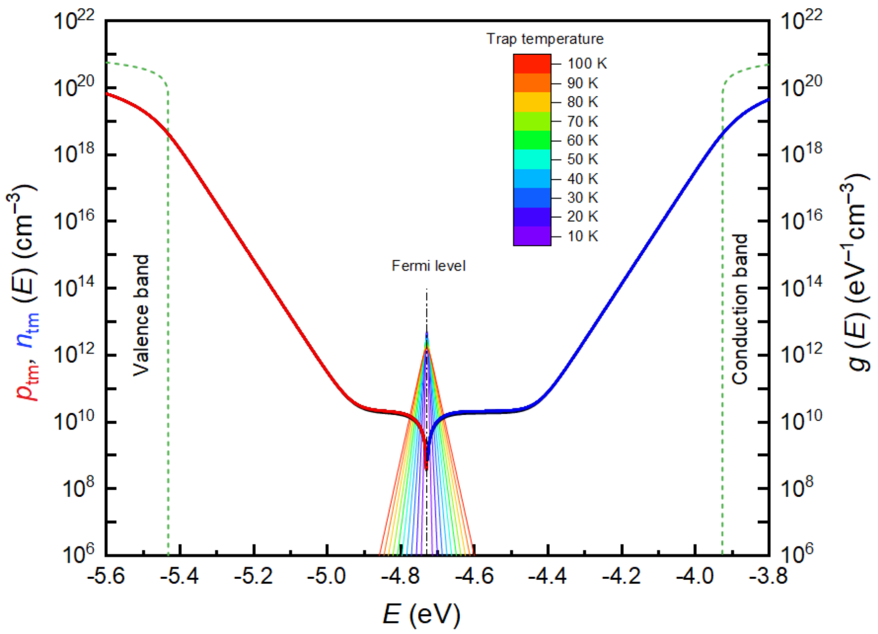


Fig. S12 The effect of temperature T_t (K) on p_{tm} and n_{tm} in the range of 10–100 K (10 K step).

The temperature of monoenergetic trap state T_t (K) does not significantly affect the trapped charge concentrations p_{tm} and n_{tm} as can be seen in Fig. S12. Overall, by applying the A-SCLC one can precisely map which charge carriers are influenced with the trap and model the charge trapping in the given material.

Supplementary Note 4: Injection and extraction model of the *J*-*V* characteristics.

Two models were used to describe the *J*-*V* characteristics of diodes based on halide perovskite materials: (i) injection model describing diode in forward bias and (ii) extraction model, describing the extraction of charge carriers under applied bias voltage. This algorithm enables the determination of the concentration of free and trapped charge carriers during injection or extraction of electric charge (electrons or holes) at a given quasi-Fermi level position. When the quasi-Fermi level shifts closer to the valence band due to injection, it means that the concentration of holes (majority carriers) increases and, conversely, the concentration of electrons (minority carriers) decreases and vice versa. In our study, the injection model is sufficient to fully describe the behavior of $MAPbBr_3$ single crystals (dark, after illumination). For $MAPbI_3$ single crystals, where hole injection barrier is observed under dark conditions (i.e. extraction mechanism dominates), the extraction diode model must be implemented (see Fig. 2b). Interestingly, it is shown that the proposed model can evaluate this phenomenon as well. All presented current-voltage characteristics were measured in the forward bias regime only (0V to 3V).

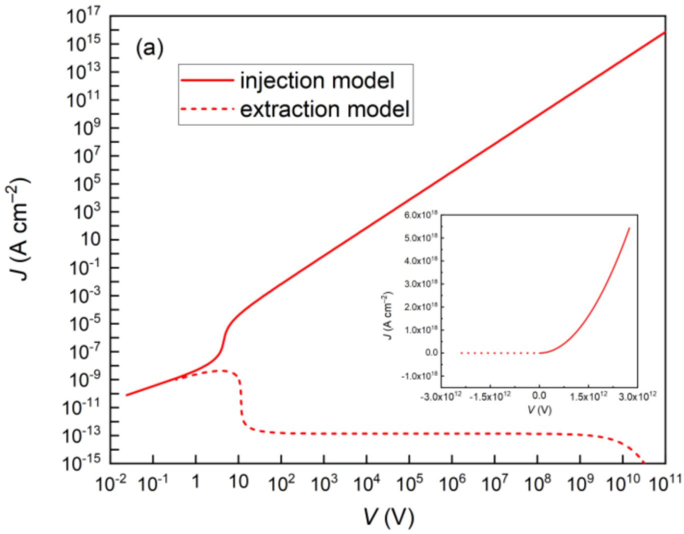


Fig. S13 Injection and extraction models of *J*-*V* characteristics in log-log (inset) and lin-lin scale plotted in broad voltage range.

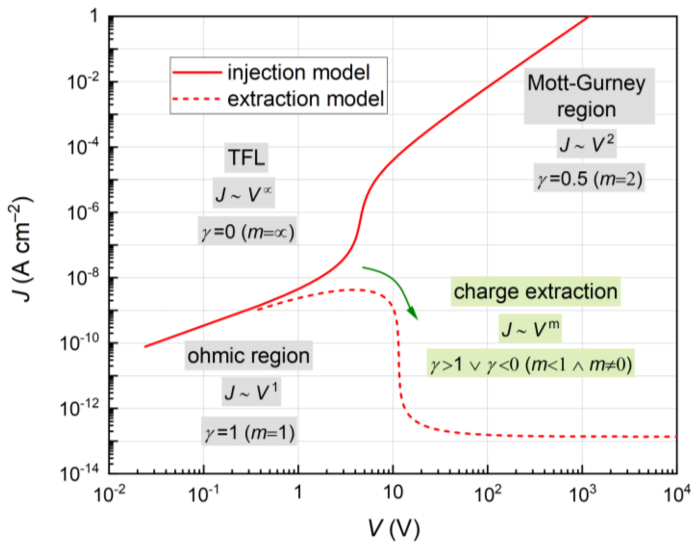


Fig. S14 Injection and extraction models of *J*-*V* characteristics in shorter range with the corresponding regions (ohmic, trap filled limit, Mott-Gurney).

Supplementary Note 5: Flexibility of the model and parameter theta (*Θ*).

The advanced-SCLC model presented in this work was applied to two semiconductor materials. For simplicity, it was assumed that these semiconductors have one monoenergetic trap state. However, advanced SCLC model is a flexible approach that enables fitting with multiple trap states. Fig. S15 shows model current-voltage characteristics for material that possesses two monoenergetic (a) trap states and their positions in the bandgap (b). All regions of the *J*-*V* are marked.

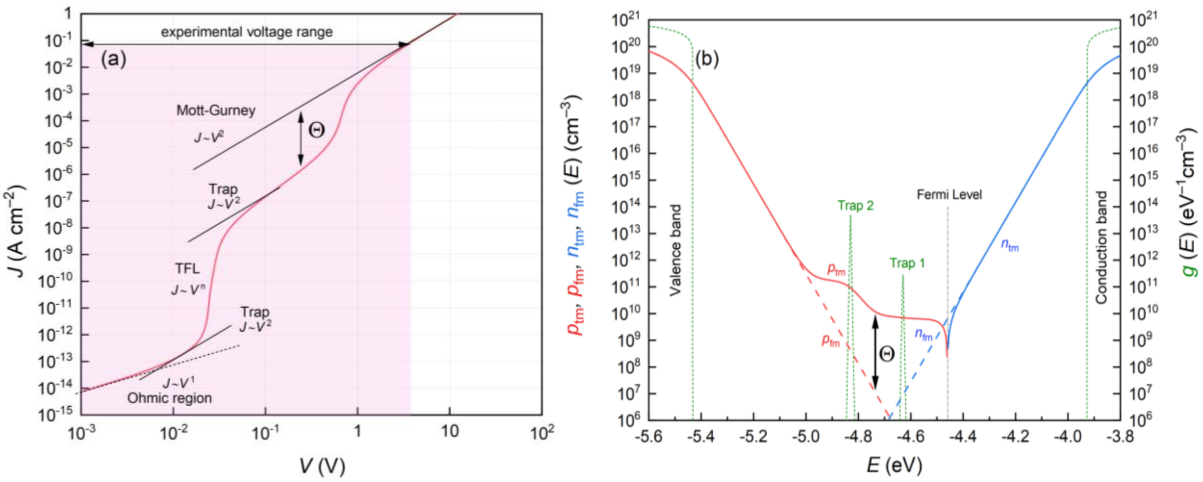


Fig. S15 (a) *J*-*V* characteristics after adding an additional trap state and (b) the position of the two trap states in the bandgap.

In addition, the figures show the meaning of parameter theta (*Θ*), described with the following relation:

$$\Theta = p_f / p_s = p_f / (p_f + p_t) \quad \text{(S12)}$$

Supplementary Note 6: Mobile ion concentration and its effect on the measured *J*-*V* curve.

Hysteresis is a well-known phenomenon observed when measuring *J*-*V* curves of halide perovskite-based devices. As many suggest, this phenomenon originates from the ionic species (i.e. transport of native mobile ions/defects) present in halide perovskites. However, it is important to realize that the number of ions affecting the charge transport is related only to the activated ions[11]. Thus, it is not surprising that ions affect the measured *J*-*V* curve and the parameters evaluated by the space charge limited currents method. So far, the effect of ions on charge transport in halide perovskites has been discussed in many publications[5],[6],[11]–[15].

In this case, the general equation of Ohm's law in differential form can be written as

$$J = e\,p\,\mu_p E + e\,n\,\mu_n\,E + eD_n\frac{\mathrm{d}n}{\mathrm{d}x} - eD_p\frac{\mathrm{d}p}{\mathrm{d}x}, \quad \text{(S13)}$$

where e is an elementary charge ($e = 1.602 \cdot 10^{-19}$ C), μ_p is the microscopic mobility for holes (μ_n for electrons), p (n) is a concentration of holes (electrons), E is an electric field and D_n and D_p are diffusion coefficients for negative charge species and positive charge species, respectively.

When activated ions affect the charge transport, it is necessary to include the diffusion elements in the modeling of the *J-V* curves, as described in the equations S13 and S15. In our case, we measured the *J-V* curves at low scan rate of 1 mV/s in forward (from 0 V to 3 V) bias mode only (we plotted all raw data points and the corresponding averaged points in Fig. S17 and Fig. S18). Thus, we consider our measurement to be a steady state. When compared with other published papers[11],[15], it is obvious that many recognized authors point out that the low scan rate completely excludes hysteresis curves, a typical after-effect of ion-related phenomena.

Moreover, if we take into account the band diagram of the studied devices C/perovskite/C and holes as majority charge carriers, then the Ohm's law equation can finally be modified to the following form (if ions are negligible):

$$J = e\mu_0 p_{\mathrm{f}} (2-\gamma)\frac{V}{L}. \tag{S14}$$

The equation S14 was used to process all results presented in this publication. However, we realize that in these materials, diffusion of ionic species is always present. Therefore, to confirm our statement about negligible ionic effect, we compared the determined diffusion coefficients for studied crystals with the results presented by Le Corre, et. al. in "Revealing Charge Carrier Mobility and Defect Densities in Metal Halide Perovskites via Space-Charge-Limited Current Measurements". In this article, authors stated that as the mobility gets higher than 10^{-4} $cm^{-2}V^{-1}s^{-1}$, the effect of ions starts to influence the *J-V*. In this case, the effect of ions can be excluded by determining the diffusion coefficients and mobility of mobile ionic species. As we have shown in our previously published paper[16], the diffusion coefficients of perovskite single crystals are: $(1.1 \pm 0.6) \times 10^{-6}$ cm^2s^{-1} for $MAPbBr_3$ and $(4 \pm 2) \times 10^{-9}$ cm^2s^{-1} for $MAPbI_3$. These values are in a good agreement with previously published results[11],[17]. From these diffusion coefficients, the mobilities can then be easily calculated (for a temperature of 300K) based on the Einstein equation

$$D = \frac{\mu k_{\mathrm{B}} T}{e}, \tag{S15}$$

where μ is the mobility, k_B is the Boltzmann's constant and T is the temperature.

The determined mobilities are 4.25×10^{-5} $cm^{-2}V^{-1}s^{-1}$ for $MAPbBr_3$, 1.39×10^{-7} $cm^{-2}V^{-1}s^{-1}$ for $MAPbI_3$ single crystals. Based on these mobilities and the magnitude of the applied electric field, it can be deduced that the influence of ions can be neglected in our *J-V* curves. However, the mobility of ions for $MAPbBr_3$ crystal is at the limit of affecting the *J-V* measurement, and thus we decided to measure the capacitance spectra and further check our hypothesis about ionic phenomena. Previously published publication on this topic suggest that two techniques can be used for determining the active mobile ions and their concentrations: (i) impedance spectroscopy[18] or (ii) capacitance transient measurements[19],[20]. Using the approach described for impedance spectroscopy (IS) measurements under external electric field (biased at 3V, see Fig. S16), we determined the concentration of activated ions.

Evaluation of IS measurement was done by calculating the derivation $-f\mathrm{d}C/\mathrm{d}f$ and from the maximum of localized peaks the characteristic frequency can be found. Subsequently, we applied the ionic Debye layer[21],[22], defined with the following equation

$$N_{\mathrm{ion}} = \frac{k_{\mathrm{B}} T \Delta C^2}{e^2 \varepsilon}. \tag{S16}$$

Using this equation, we determined the concentration of mobile ions at the interfaces of the active layer, driven by the external electric field (that is, from the excess capacitance). The ion concentrations were calculated as following: 1.65×10^9 cm^3 for $MAPbBr_3$ and 4.52×10^7 cm^3 for $MAPbI_3$. These values are in good agreement with the results presented by Shi, D. et al. and Rosenberg, J. W. et. al. in low trap-state density of halide perovskite single crystals[9],[22]. Based on the supporting evidence on ionic-related behavior in halide perovskite materials, we conclude that the influence of ions is negligible in our experiment and thus, it can be excluded from the calculation during A-SCLC evaluation (for both cases).

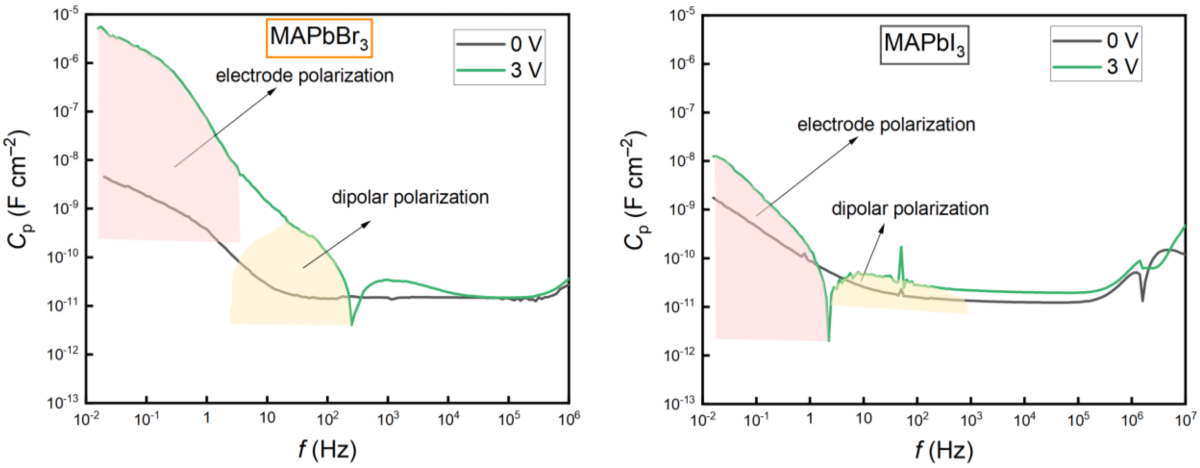


Fig. S16 Frequency dependence of capacitance for $MAPbBr_3$ and $MAPbI_3$ macroscopic single crystals measured in the dark at 0 V and biased at 3 V.

Supplementary experimental data.

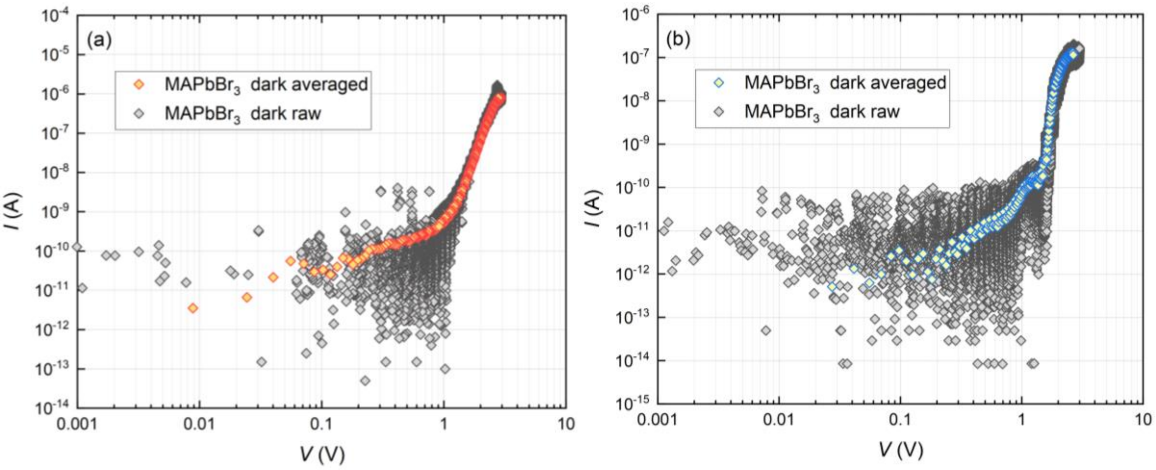


Fig. S17 Raw and averaged data of light (a) and dark (b) *I-V* characteristics of C/$MAPbBr_3$/C.

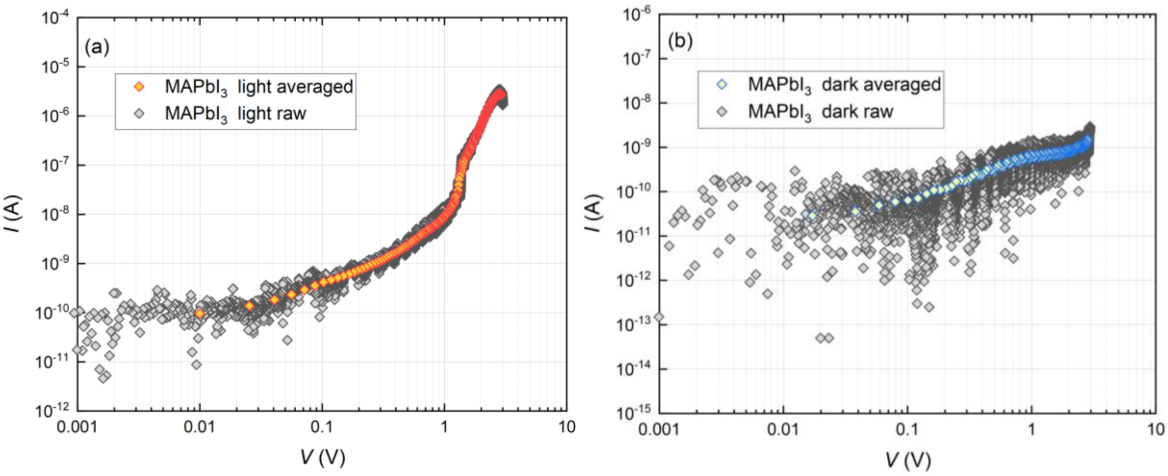


Fig. S18 Raw and averaged data of light (a) and dark (b) *I-V* characteristics of C/$MAPbI_3$/C.

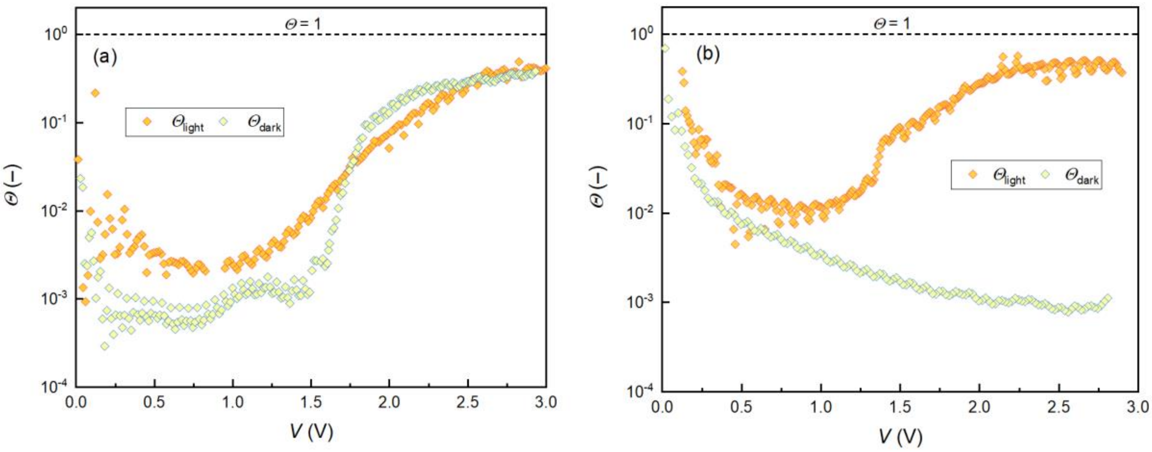
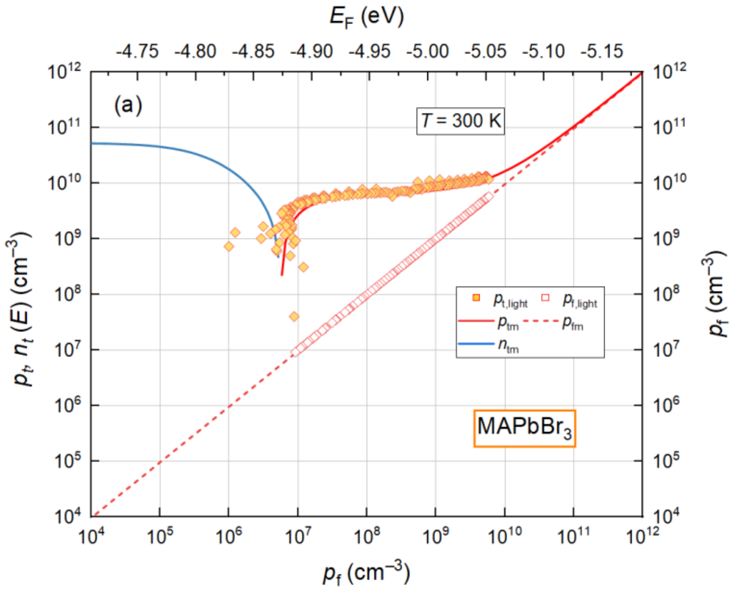


Fig. S19 Parameter theta calculated for (a) $MAPbBr_3$ and (b) $MAPbI_3$.



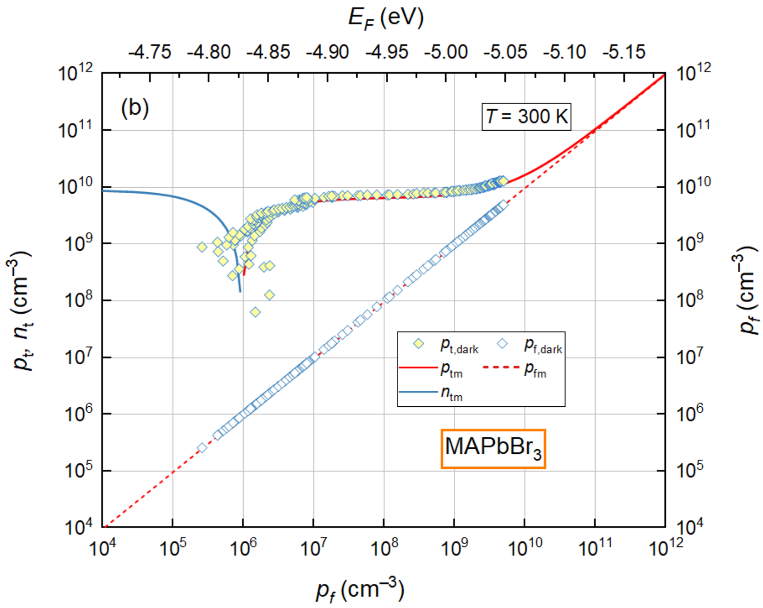


Fig. S20 Light (a) and dark (b) p_t dependences on p_f ($p_t = f(p_f)$) for $MAPbBr_3$ single crystal.

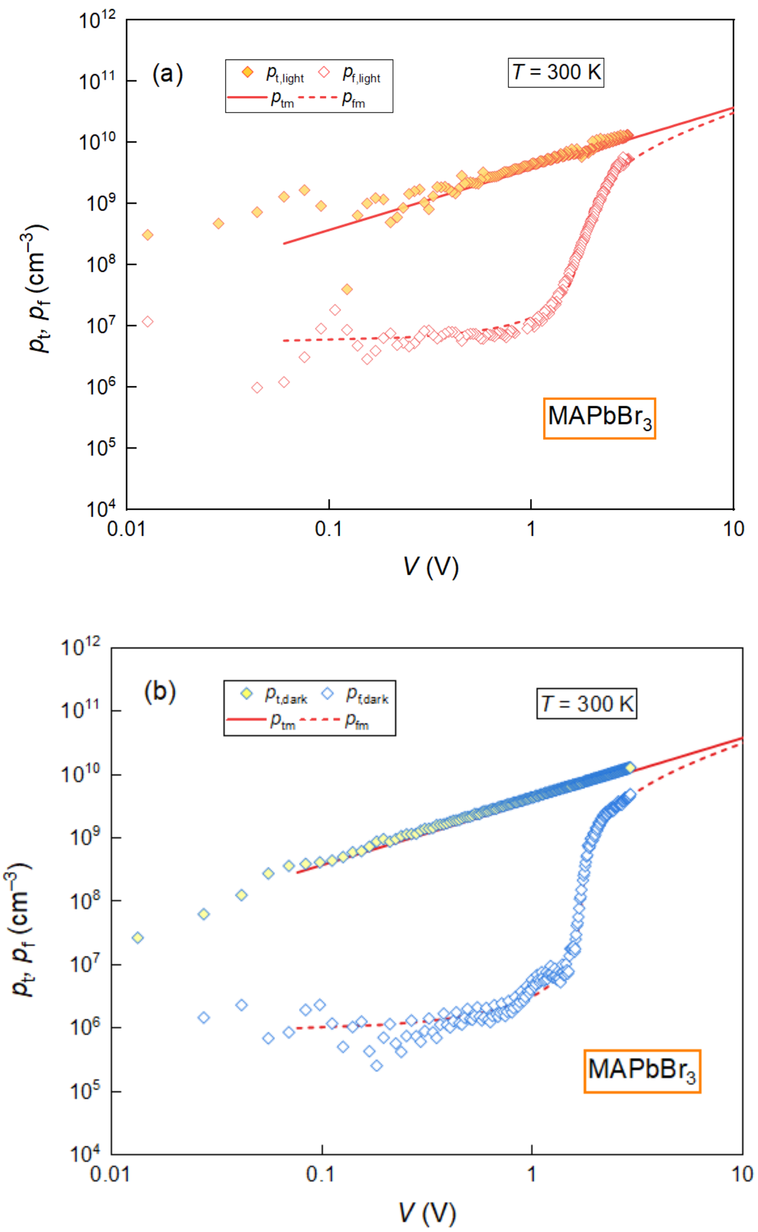


Fig. S21 Light (a) and dark (b) voltage dependencies of p_t and p_f for $MAPbBr_3$ single crystal.

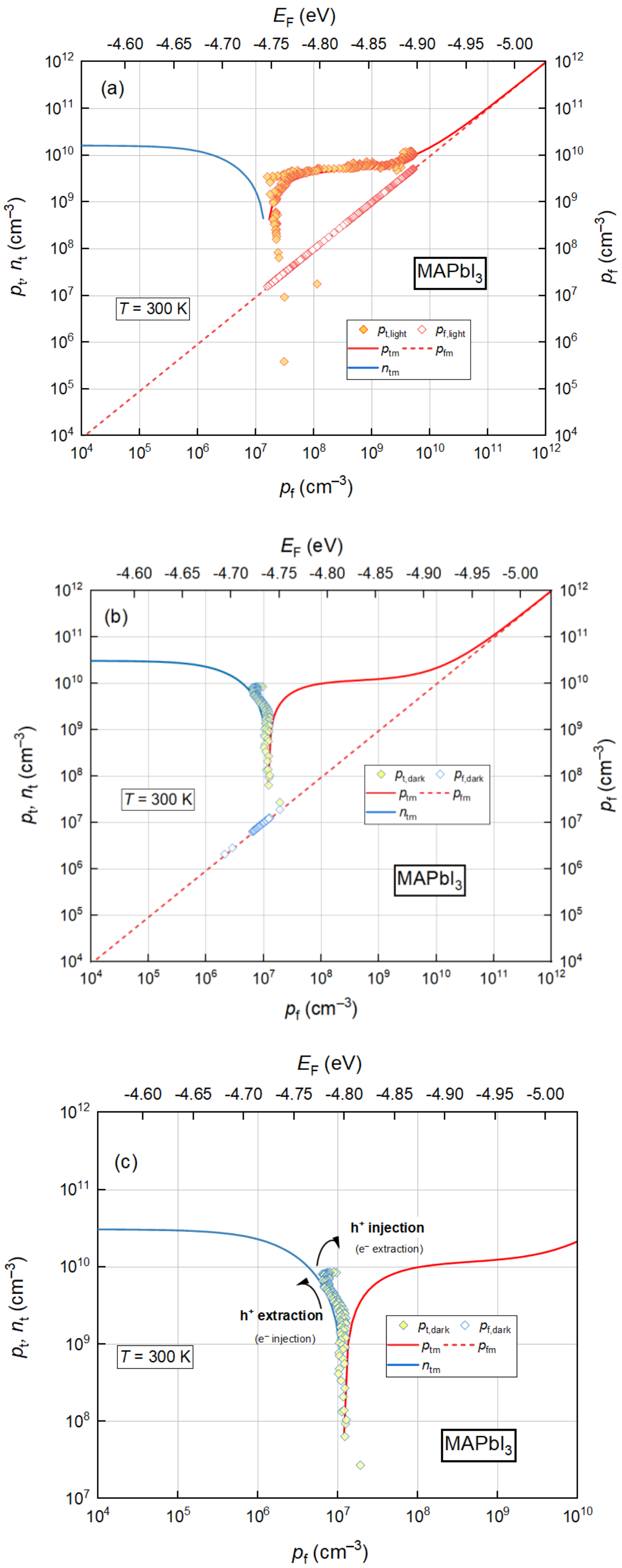
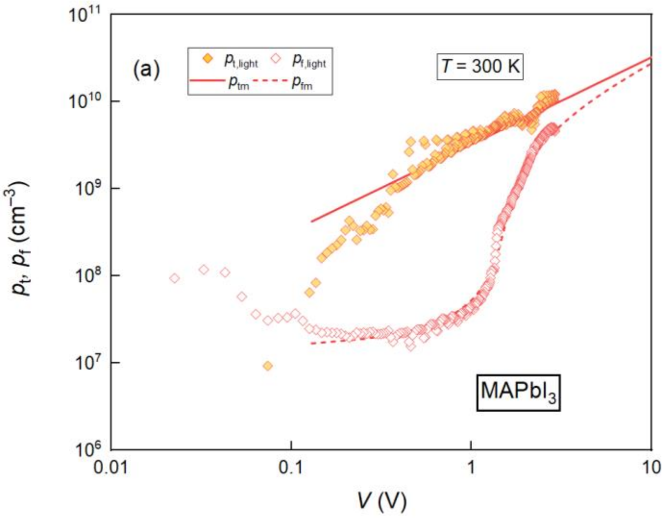


Fig. S22 Light (a) and dark (b,c) p_t dependences on p_f ($p_t = f(p_f)$) for $MAPbI_3$ single crystal.



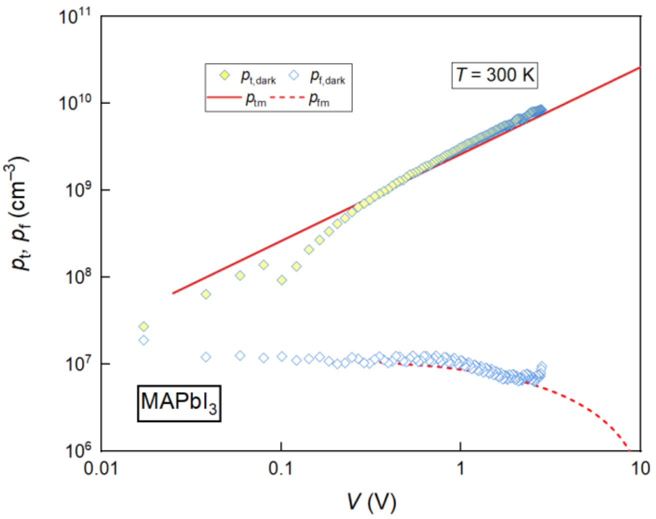


Fig. S23 Light (a) and dark (b) voltage dependencies of p_t and p_f for $MAPbI_3$ single crystal.

Table S3 Previously published SCLC parameters for $MAPbBr_3$ and $MAPbI_3$ single crystals.

Perovskite	**Architecture**	***L* (mm)**	**$p_t(n_t)\times10^{10}$ (cm^{-3})**		**μ ($cm^2V^{-1}s^{-1}$)**		**Reference**
$MAPbBr_3$	$Au/MAPbBr_3/Au$	2.32	3		24		Saidaminov et. al.[23]
	$C/MAPbBr_3/C$	~1	6.2		17.8		Pospisil et. al.[24]
	$Au/MAPbBr_3/Au$*	1.6	0.58		38		Shi et. al.[9]
	$Au/MAPbBr_3/Au$	0.1	1 300		13		Le Corre et. al.[6]
		0.16	280		/		Duijnstee et. al.[5]
		2.5	0.52		81		Liu et. al.[26]
	$C/MAPbBr_3/C$	0.80	dark 0.76	light 0.88	dark 50	light 160	this work
$MAPbI_3$	$Au/MAPbI_3/Au$	2.49	1.4		67.2		Saidaminov et. al.[23]
		~3	3.6		164		Dong et. al.[27]
	$Au/MAPbBr_3/Au$*	1.6	3.3		/		Shi et. al.[9]
	$Au/MAPbI_3/Au$	1.2	1.8		117		Huang et. al.[25]
	$C/MAPbI_3/C$	1.05	dark 1.15	light 0.86	dark 90	light 230	this work

*Shi et. al. sample architecture: $Ag/Au/MoO_3$/pvsk/MoO_3/Au/Ag.

Table S4 All parameters used for modeling the *J-V* characteristics of $MAPbBr_3$ and $MAPbI_3$.

Parameters	$MAPbBr_3$		$MAPbI_3$	
Temperature (T)	300 K			
Voltage range (V_{min}–V_{max})	0–3 V			
Voltage scan rate (V_{scan})	1 mV s^{-1}			
Thickness (L)	0.80 mm		1.05 mm	
Area (A)	7.70 mm^2		20.10 mm^2	
Relative permittivity (ε_r)	25.5[20]		32[20]	
Effective electron mass (m^*/m_e)	0.32[1]		0.35[2]	
Conductive band energy (E_c)	–3.36 eV		–3.93 eV	
Valence band energy (E_v)	–5.58 eV		–5.43 eV	
Bandgap energy (E_g)	2.20 eV		1.50 eV	
Concentration of delocalized states ($N_{c,v}$)	4.52×10^{16} cm^{-3}		5.17×10^{10} cm^{-3}	
Concentration of localized states (N_t)	dark	light	dark	light
	3.03×10^{10} cm^{-3}	3.54×10^{10} cm^{-3}	4.60×10^{10} cm^{-3}	3.45×10^{10} cm^{-3}
Concentration of trapped charge carriers (p_t)	dark	light	dark	light
	7.58×10^{9} cm^{-3}	8.84×10^{9} cm^{-3}	1.15×10^{10} cm^{-3}	8.63×10^{9} cm^{-3}
Microscopic mobility (μ_0)	dark	light	dark	light
	50 $cm^2V^{-1}s^{-1}$	160 $cm^2V^{-1}s^{-1}$	90 $cm^2V^{-1}s^{-1}$	230 $cm^2V^{-1}s^{-1}$
Trap position (E_g)	–4.768 eV		–4.715 eV	
Trap temperature (T_t)	dark	light	dark	light
	8	40	40	95

Supplementary references

[1] Borges-Martínez, M., et al. Computational Design and Properties Elucidation of New $(FAPbI_3)_{1-X-Y}(MAPbBr_3)_Y(CsPbBr_3)_X$ Photoactive Systems for their Application in Perovskite Solar Cells. *Materials Today Communications* 34, 2023, p. 105324, doi: 10.1016/j.mtcomm.2023.105324.

[2] Yang, Jin-Peng, et al. Band Dispersion and Hole Effective Mass of Methylammonium Lead Iodide Perovskite. *Solar Rrl*. 2 (10), 2018, doi: 10.1002/solr.201800132.

[3] Neamen, Donald A. *Semiconductor Physics and Devices: Basic Principles*. 4th ed. New York, NY: McGraw-Hill, 2003, ISBN: 978-0-07-352958-5.

[4] Jin, H. et al. It's a trap! On the Nature of Localised States and Charge Trapping in Lead Halide Perovskites. *Materials Horizons* 7, 397-410 (2020), doi: 10.1039/C9MH00500E.

[5] Duijnstee, E. A.; Ball, J. M.; Le Corre, V. M.; Koster, L. J. A.; Snaith, H. J.; Lim, J. Toward Understanding Space-Charge Limited Current Measurements on Metal Halide Perovskites. ACS Energy Letters 2020, 5 (2), 376-384.

[6] Le Corre, V. M.; Duijnstee, E. A.; El Tambouli, O.; Ball, J. M.; Snaith, H. J.; Lim, J.; Koster, L. J. A. Revealing Charge Carrier Mobility and Defect Densities in Metal Halide Perovskites Via Space-Charge-Limited Current Measurements. ACS Energy Letters 2021, 6 (3), 1087-1094.

[7] Sajedi Alvar, M., Blom, P. W. M. & Wetzelaer, G. -J. A. H. Space-charge-limited electron and hole currents in hybrid organic-inorganic perovskites. Nature Communications 11, (2020) doi:10.1038/s41467-020-17868-0.

[8] Tan, S. et al. Inorganic Ammonium Halide Additive Strategy for Highly Efficient and Stable $CsPbI_3$ Perovskite Solar Cells. Advanced Functional Materials 31, (2021) doi:10.1002/adfm.202010813.

[9] Shi, D. et al. Low Trap-State Density and Long Carrier Diffusion in Organolead Trihalide Perovskite Single Crystals. Science 347, 519-522 (2015) doi:10.1126/science.aaa2725.

[10] Saidaminov, M. I. et al. Planar-integrated single-crystalline perovskite photodetectors. Nature Communications 6, (2015) doi:10.1038/ncomms9724.

[11] Peng, W. et al. Quantification of Ionic Diffusion in Lead Halide Perovskite Single Crystals. ACS Energy Letters 3, 1477-1481 (2018) doi:10.1021/acsenergylett.8b00641.

[12] García-Batlle, M. et al. Effective Ion Mobility and Long-Term Dark Current of Metal Halide Perovskites with Different Crystallinities and Compositions. Advanced Photonics Research 3, (2022) doi:10.1002/adpr.202200136.

[13] Thiesbrummel, J. et al. Ion-induced Field Screening as a Dominant Factor in Perovskite Solar Cell Operational Stability. Nature Energy 9, 664-676 (2024) doi:10.1038/s41560-024-01487-w.

[14] Sakhatskyi, K. et al. Assessing the Drawbacks and Benefits of Ion Migration in Lead Halide Perovskites. ACS Energy Letters 7, 3401-3414 (2022) doi:10.1021/acsenergylett.2c01663.

[15] Tress, W. et al. Understanding the Rate-Dependent J–V Hysteresis, Slow Time Component, and Aging in $CH_3NH_3PbI_3$ Perovskite Solar Cells: the Role of a Compensated Electric Field. Energy & Environmental Science 8, 995-1004 (2015) doi:10.1039/C4EE03664F.

[16] Pospisil, J., Marackova, L., Zmeskal, O. & Kovalenko, A. Universal Approach for Diffusion Quantification Applied to Lead Halide Perovskite Single Crystals. Applied Physics A 129, (2023) doi:10.1007/s00339-023-06398-3.

[17] Hoque, M. N. F. et al. Ionic and Optical Properties of Methylammonium Lead Iodide Perovskite across the Tetragonal–Cubic Structural Phase Transition. ChemSusChem 9, 2692-2698 (2016) doi:10.1002/cssc.201600949.

[18] Almora, O. et al. Capacitive Dark Currents, Hysteresis, and Electrode Polarization in Lead Halide Perovskite Solar Cells. The Journal of Physical Chemistry Letters 6, 1645-1652 (2015) doi:10.1021/acs.jpclett.5b00480.

[19] McGovern, L., Futscher, M. H., Muscarella, L. A. & Ehrler, B. Understanding the Stability of $MAPbBr_3$ versus $MAPbI_3$: Suppression of Methylammonium Migration and Reduction of Halide Migration. The Journal of Physical Chemistry Letters 11, 7127-7132 (2020) doi:10.1021/acs.jpclett.0c01822.

[20] Futscher, M. H. et al. Quantification of ion migration in $CH_3NH_3PbI_3$ perovskite solar cells by transient capacitance measurements. Materials Horizons 6, 1497-1503 (2019) doi:10.1039/C9MH00445A.

[21] Reichert, S. et al. Probing the ionic defect landscape in halide perovskite solar cells. Nature Communications 11, (2020) doi:10.1038/s41467-020-19769-8.

[22] Rosenberg, J. W., Legodi, M. J., Rakita, Y., Cahen, D. & Diale, M. Laplace current deep level transient spectroscopy measurements of defect states in methylammonium lead bromide single crystals. Journal of Applied Physics 122, (2017) doi:10.1063/1.4995970.

[23] Saidaminov, M. I. et al. High-quality bulk hybrid perovskite single crystals within minutes by inverse temperature crystallization. Nature Communications 6, (2015) doi:10.1038/ncomms8586.

[24] Pospisil, J. et al. Density of bulk trap states of hybrid lead halide perovskite single crystals: temperature modulated space-charge-limited-currents. Scientific Reports 9, (2019) doi:10.1038/s41598-019-40139-y.

[25] Huang, Y. et al. A-site Cation Engineering for Highly Efficient $MAPbI_3$ Single-Crystal X-ray Detector. Angewandte Chemie International Edition 58, 17834-17842 (2019) doi:10.1002/anie.201911281.

[26] Liu, Y. et al. A 1300 mm^2 Ultrahigh-Performance Digital Imaging Assembly using High-Quality Perovskite Single Crystals. Advanced Materials 30, (2018) doi:10.1002/adma.201707314.

[27] Dong, Q. et al. Electron-hole diffusion lengths 175 μm in solution-grown $CH_3NH_3PbI_3$ Single Crystals. Science 347, 967-970 (2015) doi:10.1126/science.aaa5760.